

A. **Wilson's Theorem.** Next a result of 18th century mathematician John Wilson.

**Wilson's Theorem.** If  $p$  is prime, then

**Proof.** Let  $p$  be prime. First, if  $p = 2$ , it is obvious that

Next, suppose, that  $p$  is an odd prime.

Consider the list of integers:

$1, 2, \dots, p-1$

Each integer  $a$  in this list is  $\perp p$ , and thus has an inverse, and

**Lemma.** In this list, only  $a = 1$  and  $a = p-1$  have the property  $a \equiv a^{-1} \pmod{p}$ .

**Proof.** Multiplying  $a \equiv a^{-1} \pmod{p}$  by  $a$ , this congruence is equivalent to

Because  $p$  is prime, it must divide one of:

Hence:  $a \equiv$

and thus the only  $a$  in the list  $1, \dots, p-1$  that satisfy  $a \equiv a^{-1} \pmod{p}$  are:

□

According to the lemma, besides  $1$  and  $p-1$ , the integers in the list come in pairs

As an example, consider  $p = 7$ , in which case  $1, 2, 3, 4, 5, 6$  are paired off as

Hence, when we multiply all entries in the list together, only  $1$  and  $p-1$  survive.

$(p-1)! \equiv$

□

**Example 1.** Find the remainder when  $124!$  is divided by  $127$ .

**Solution.** First, it can be verified that  $127$  is prime.

So Wilson's theorem applies and says

**Example 2.** Prove that if  $n$  is composite, then

$$(n-1)! \not\equiv -1 \pmod{n}$$

Therefore, we can turn Wilson's Theorem into an if and only if statement.

**Wilson:**

**Proof.** We can be more specific. Let's prove that if  $n$  is composite, then

except when  $n = 4$ , where:

It turns out we need to break into cases on whether  $n$  is a square of a prime.

**Case 1.** Suppose  $n$  is composite and not a square of a prime. Then

Hence  $a$  and  $b$  appear separately in the list of integers from  $1$  to  $n-1$ .

Thus:

**Case 2.** Suppose  $n = p^2$  for some prime  $p$  greater than  $2$ .

$$2p$$

Hence both  $p$  and  $2p$  appear in the list of factors terms  $1$  to  $n-1 = p^2-1$ .

Thus:

□

**B. Roots of Negative Unity.** Let  $p$  be prime and consider the congruence

Unity is used to mean one.

$$x^2 \equiv -1 \pmod{p}$$

Is it solvable? In other words, does  $-1$  have a square-root modulo  $p$ ?

A quick check shows the following.

$$x^2 \equiv -1 \pmod{3} \text{ has:}$$

while

$$x^2 \equiv -1 \pmod{5} \text{ has:}$$

**Theorem.** Let  $p$  be an odd prime.

The congruence  $x^2 \equiv -1 \pmod{p}$  has a solution if and only if:

**Proof.**  $\implies$  : Assume  $x^2 \equiv -1 \pmod{p}$  has a solution  $x$ .

Towards a contradiction, assume  $p$  has form:

By Fermat

$$1 \equiv x^{p-1} \equiv$$

This is impossible, hence  $p$  must be of the form:

$\Leftarrow$  : Assume  $p$  has form:

We will show that

$$x =$$

squares to  $-1$ .

Take the list  $1, \dots, p-1$  and split it in half, using that  $p-1 =$

Therefore

$$(p-1)! \equiv$$

But by Wilson's Theorem:

□